Introduction to Quadratic Equations - Nathaniel Juan Put

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Student

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Topic: Introduction to Quadratic Equations

Status: completed

Lesson Plan

Learning Objectives

- Construct quadratic equations in standard form ($ax^2 + bx + c = 0$) from various mathematical contexts.
- Calculate the roots of quadratic equations using the factoring method.
- Solve for the x-intercepts of a quadratic function by interpreting its graphical representation.
- Analyze the relationship between the algebraic factors of a quadratic and its zeros on a coordinate plane.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Nathaniel! Before we dive into today's new challenge, let's sharpen your tools with a quick brain warm-up. We're going to revisit our work on logarithms. Look at the virtual whiteboard. I've posted four quick challenges. For the first two, use the Change of Base formula. For the next two, apply your log properties. Type your final answers in the chat as soon as you get them!

Activities:

- *Solve: Change of base for log_8(32).*
- *Solve: Evaluate log_3(10) using common logs (log 10 / log 3).*
- *Solve: Simplify log(x^3) - log(x).*
- *Solve: Expand log(5ab^2).*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Great work on those logs. Today, we transition to Quadratic Equations. Imagine you are designing a suspension bridge or calculating the trajectory of a rocket. These paths aren't straight lines; they are curves called parabolas. The math behind these curves is the Quadratic Equation. Today, we move from linear 'flat' math to 'curved' math that defines the physical world around us.

Activities:

- *Visual showcase of the Gateway Arch and a basketball shot trajectory.*
- *Discussion on the 'u-shape' vs 'straight-line' geometry.*

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Before we factor quadratics, we need to remember the Zero Product Property. If I tell you that two numbers multiplied together equal zero, what do we know about one of those numbers? Exactly, at least one must be zero. Let's practice: if $(x-3)(x+2) = 0$, what are the possible values for x? Keep this in mind, it's our secret weapon today.

Activities:

- *Mental math: Solving $(x-5)(x+5)=0$.*
- *Reviewing distributive property (FOIL) in reverse.*

4. Problem Introduction (10 minutes)

Teacher Script:

Let's look at the Standard Form: $ax^2 + bx + c = 0$. Notice the 'squared' term; that's what makes it quadratic. Our goal is to find the 'roots'—the values of x that make the equation true. We can do this by 'un-multiplying' the equation into two binomials. Let's look at $x^2 + 5x + 6 = 0$. We need two numbers that multiply to 6 and add to 5. What do you think?

Activities:

- *Step-by-step breakdown of identifying a, b, and c.*
- *Introduction to the 'Diamond Method' for finding factors.*

5. Guided Practice (15 minutes)

Teacher Script:

Okay Nathaniel, let's try a harder one together: $x^2 - 2x - 8 = 0$. First, identify our targets. What is our product target (c)? What is our sum target (b)? Now, let's list factors of -8. Since the product is negative, one factor must be positive and one negative. Walk me through your logic as you test them out on the board.

Activities:

- *Collaborative solving of x^2 - 2x - 8 = 0.*
- *Graphing the equation using Desmos to see where the curve touches the x-axis.*

6. Independent Practice (15 minutes)

Teacher Script:

You're handling the logic perfectly. Now, I have three challenges for you. The first is a standard factoring problem. The second has a negative 'a' value—try factoring out -1 first! The third is a graph: I want you to look at the x-intercepts and tell me what the equation's factors must be. Go for it!

Activities:

- *Task 1: Solve $x^2 + 7x + 12 = 0$.*
- *Task 2: Solve $2x^2 - 10x + 12 = 0$ (Requires GCF first).*
- *Task 3: Identify roots from a pre-drawn graph on the whiteboard.*

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Brilliant session, Nathaniel. You've mastered how to take a complex curve and break it down into simple linear factors. Remember, the roots are just the points where our path hits the ground. Before you go, please log into your dashboard at https://learn.algonova.id/login to access your homework and review today's notes. Any questions on the factoring logic today?

Activities:

- *Summary of Standard Form vs Factored Form.*
- *Login reminder.*

Homework

Medium Questions:

1. Solve by factoring: $x^2 + 8x + 15 = 0$
2. Solve by factoring: $x^2 - 9x + 20 = 0$
3. Identify a, b, and c in the equation: $5x^2 - 2x = -7$
4. Solve by factoring: $x^2 + 2x - 24 = 0$

Hard Questions:

1. Solve by factoring: $2x^2 + 7x + 3 = 0$
2. Solve by factoring: $3x^2 - 10x + 8 = 0$
3. A parabola crosses the x-axis at $x = 4$ and $x = -3$. Write the quadratic equation in standard form.
4. Solve by factoring: $-x^2 + 5x + 14 = 0$ (Hint: factor out -1 first)
5. Determine the roots of $x^2 - 16 = 0$ by factoring as a difference of squares.
6. Solve for x: $x(x - 5) = 14$

Answer Key:

Q1: $(x+3)(x+5)=0 \rightarrow x=-3, -5$
Q2: $(x-4)(x-5)=0 \rightarrow x=4, 5$
Q3: $5x^2 - 2x + 7 = 0 \rightarrow a=5, b=-2, c=7$
Q4: $(x+6)(x-4)=0 \rightarrow x=-6, 4$
Q5: $(2x+1)(x+3)=0 \rightarrow x=-1/2, -3$
Q6: $(3x-4)(x-2)=0 \rightarrow x=4/3, 2$
Q7: $(x-4)(x+3) = x^2 - x - 12 = 0$
Q8: $-(x^2-5x-14)=0 \rightarrow -(x-7)(x+2)=0 \rightarrow x=7, -2$
Q9: $(x-4)(x+4)=0 \rightarrow x=4, -4$
Q10: $x^2-5x-14=0 \rightarrow (x-7)(x+2)=0 \rightarrow x=7, -2$
REVIEW: Change of Base - $\log_8(32) = \log(32)/\log(8) = 5/3$
REVIEW: Change of Base - $\log_2(7)$ hard: $\log(7)/\log(2)$ approx 2.807
REVIEW: Properties of Logs - $\log(x^3) - \log(x) = \log(x^2) = 2\log(x)$
REVIEW: Properties of Logs - Simplify $2\log(5) + \log(4) = \log(25*4) = \log(100) = 2$

Parent Reminder

Math Progress Update: Nathaniel Juan Putra Winata Eng

Today Nathaniel successfully transitioned into Quadratic Equations! We explored how these equations represent curves in physics and architecture. He practiced solving them by factoring and identifying roots on a graph. Please remind him to complete the homework on the dashboard to reinforce these new concepts.